

# MICROSOFT EXCEL 365

## Basic to Intermediate Level

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### *Participant Training Notes*

Hands-On Corporate Training Program | 2 Days | 11 Modules

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# DAY 1 — MODULE 1: Introduction to Microsoft Excel 365

## 1.1 What is Microsoft Excel?

Microsoft Excel is a powerful spreadsheet application developed by Microsoft. It is used to organize, analyze, calculate, and visualize data. Excel is widely used across industries — from finance and HR to operations and marketing — making it one of the most essential workplace skills globally.

Excel 365 is the cloud-connected version included in the Microsoft 365 subscription. It receives continuous updates and includes powerful new features like XLOOKUP, LET, LAMBDA, and AI-powered Microsoft Copilot.

## 1.2 What's New in Excel 365?

| Feature           | Description                                     | Why It Matters                                      |
|-------------------|-------------------------------------------------|-----------------------------------------------------|
| XLOOKUP           | Modern replacement for VLOOKUP                  | Searches any direction, no column numbers needed    |
| LET Function      | Assign names to calculations inside a formula   | Makes long formulas cleaner and faster              |
| LAMBDA Function   | Create your own custom reusable functions       | Advanced automation without VBA                     |
| Dynamic Arrays    | Formulas that spill results into multiple cells | FILTER, SORT, UNIQUE, SEQUENCE                      |
| Microsoft Copilot | AI assistant built into Excel                   | Generate formulas and insights using plain language |

## 1.3 The Excel 365 Interface

 [Screenshot: Excel 365 Full Interface — Label all key areas]

### Key Interface Areas:

- Title Bar — Shows the workbook name. Includes AutoSave toggle (for OneDrive files) and Account info.
- Quick Access Toolbar (QAT) — Customizable toolbar at the top. Add your most-used commands here (e.g., Save, Undo, Print Preview).
- Ribbon — The main toolbar area organized into tabs: Home, Insert, Page Layout, Formulas, Data, Review, View.
- Name Box — Displays the current cell address (e.g., A1). You can type a cell address here and press Enter to navigate directly.
- Formula Bar — Displays the content of the selected cell — a value, text, or formula. Always edit formulas here.
- Column Headers (A, B, C...) — Letters identifying each column. Excel supports up to Column XFD (16,384 columns).
- Row Numbers (1, 2, 3...) — Numbers identifying each row. Excel supports up to Row 1,048,576.

- **Cell** — The intersection of a row and column. Each cell has a unique address, e.g., B5 = Column B, Row 5.
- **Sheet Tabs** — Navigate between worksheets. Right-click a tab to rename, move, copy, or colour-code it.
- **Status Bar** — The bar at the bottom. Right-click it to show Sum, Average, Count, Min, Max of selected cells instantly.

 **PRO TIP:** Press Ctrl+End to jump to the last used cell in your worksheet. This is useful to check the actual size of your dataset.

## 1.4 Ribbon Tabs Overview

| Tab         | What You'll Use It For                                                         |
|-------------|--------------------------------------------------------------------------------|
| Home        | Formatting, copy/paste, number formats, conditional formatting, find & replace |
| Insert      | Tables, Pivot Tables, charts, images, shapes, headers, hyperlinks              |
| Page Layout | Print settings, margins, orientation, page breaks, gridlines                   |
| Formulas    | Function Library, Named Ranges, error checking, formula auditing               |
| Data        | Sorting, filtering, data validation, removing duplicates, Power Query          |
| Review      | Spell check, comments, track changes, protect workbook/sheet                   |
| View        | Freeze panes, split window, zoom, show/hide gridlines, macros                  |

## 1.5 Essential Keyboard Shortcuts

| Shortcut         | Action              | Shortcut         | Action                    |
|------------------|---------------------|------------------|---------------------------|
| Ctrl + N         | New Workbook        | Ctrl + Z         | Undo                      |
| Ctrl + O         | Open File           | Ctrl + Y         | Redo                      |
| Ctrl + S         | Save                | Ctrl + C / V / X | Copy / Paste / Cut        |
| Ctrl + P         | Print               | Ctrl + F         | Find                      |
| Ctrl + Home      | Go to cell A1       | Ctrl + H         | Find & Replace            |
| Ctrl + End       | Last used cell      | Ctrl + Shift + L | Toggle AutoFilter         |
| F2               | Edit selected cell  | Alt + Enter      | New line within a cell    |
| Ctrl + 1         | Format Cells dialog | Ctrl + Shift + + | Insert row/column         |
| Ctrl + -         | Delete row/column   | Ctrl + D         | Fill Down                 |
| Ctrl + Shift + ~ | Show all formulas   | F4               | Toggle absolute reference |



## **EXERCISE: Module 1 — Explore the Excel 365 Interface**

1. Open Microsoft Excel 365 and identify each interface area (Name Box, Formula Bar, Ribbon, Sheet Tabs, Status Bar).
2. Add 'Save', 'New', and 'Print Preview' to the Quick Access Toolbar (right-click the QAT area).
3. Click on cell D10. Check the Name Box — it should show 'D10'. Press Ctrl+Home to return to A1.
4. Navigate using Ctrl+Arrow keys to understand how Excel moves through data.
5. Press Ctrl+Shift+~ on a sample worksheet to see all formulas displayed at once.

## MODULE 2: Working with Workbooks & Data Entry

### 2.1 Creating, Saving & Opening Workbooks

| Action                      | Method 1            | Method 2                     |
|-----------------------------|---------------------|------------------------------|
| Create New                  | Ctrl + N            | File → New → Blank Workbook  |
| Open Existing               | Ctrl + O            | File → Open → Browse         |
| Save                        | Ctrl + S            | File → Save                  |
| Save As (new name/location) | F12                 | File → Save As               |
| AutoSave (OneDrive)         | Toggle in Title Bar | Requires OneDrive/SharePoint |

 **NOTE:** Always save your file in .xlsx format for compatibility. Save as .xlsm only if your file contains macros.

### 2.2 Worksheet Management

#### Working with Sheet Tabs:

- Rename Sheet — Double-click the tab name and type a new name. Use short, descriptive names.
- Add New Sheet — Click the + icon next to the last sheet tab. Shortcut: Shift + F11.
- Delete Sheet — Right-click the tab → Delete. Warning: This is permanent and cannot be undone.
- Move / Copy Sheet — Right-click → Move or Copy. Check 'Create a copy' to duplicate.
- Colour a Tab — Right-click → Tab Color. Use colors to organize (e.g., green = complete, red = pending).
- Hide / Unhide Sheet — Right-click → Hide. To unhide: right-click any tab → Unhide.

### 2.3 Data Entry Best Practices

- Always put column headers in Row 1 — Use descriptive headers like 'Employee Name', 'Salary (RM)', 'Date Joined'.
- One type of data per column — Do not mix numbers and text in the same column.
- Avoid merged cells in data tables — Merging breaks sorting and filtering. Use 'Center Across Selection' instead.
- No blank rows or columns within your data — Blank rows confuse Excel's sorting and Pivot Table features.
- Use consistent date formats — Stick to DD/MM/YYYY or YYYY-MM-DD. Inconsistency causes calculation errors.

### 2.4 AutoFill

AutoFill is a smart feature that extends data or patterns automatically. To use it, enter your data in a cell (or first 2 cells for a pattern), then drag the small square (fill handle) at the bottom-right corner of the cell.

| What You Type    | AutoFill Produces                 |
|------------------|-----------------------------------|
| Monday           | Tuesday, Wednesday, Thursday...   |
| Jan              | Feb, Mar, Apr...                  |
| January          | February, March, April...         |
| Q1               | Q2, Q3, Q4                        |
| 1, 2 (two cells) | 3, 4, 5, 6...                     |
| 2, 4 (two cells) | 6, 8, 10, 12... (detects pattern) |
| EMP-001          | EMP-002, EMP-003...               |

## 2.5 Flash Fill

Flash Fill (Ctrl + E) automatically detects a pattern you are creating and fills the rest of the column for you. It is particularly useful for:

- Splitting full names into first and last names
- Combining first and last name into one column
- Extracting parts of text (e.g., extracting city from an address)
- Reformatting phone numbers or dates

 [Screenshot: Flash Fill — Before and After: extracting first names from full name column]

 **HOW TO USE FLASH FILL:** Type the first example in the cell next to your data. Start typing the second example — Excel will suggest the rest. Press Enter to accept, or use Ctrl+E to trigger manually.

## 2.6 Fill Series

Fill Series lets you create precise sequences. Go to Home → Editing → Fill → Series.

| Series Type    | Example                           | Use Case                   |
|----------------|-----------------------------------|----------------------------|
| Linear         | 1, 2, 3, 4, 5                     | Sequential numbering       |
| Growth         | 1, 2, 4, 8, 16                    | Exponential projections    |
| Date — Day     | 01/01/2024, 02/01/2024...         | Daily reports              |
| Date — Weekday | Mon, Tue, Wed... (skips weekends) | Work schedule planning     |
| Date — Month   | Jan 2024, Feb 2024...             | Monthly budget sheets      |
| AutoFill       | Custom Lists defined by you       | Department or region lists |

## 2.7 Custom Lists

Custom Lists allow you to save your own recurring sequences for AutoFill. Useful for department names, product categories, branch names, etc.

1. Go to File → Options → Advanced
2. Scroll down to the General section
3. Click 'Edit Custom Lists...'
4. In the List entries box, type your items (one per line)
5. Click Add, then OK

### EXERCISE: Module 2 — Build a Monthly Staff Report

1. Create a new workbook with three sheets: 'Staff List', 'January', 'February'.
2. In the Staff List sheet, enter headers: Employee ID, Name, Department, Designation, Date Joined, Basic Salary.
3. Use AutoFill to generate Employee IDs from EMP-001 to EMP-020.
4. In the January sheet, use Flash Fill to extract First Names from a 'Full Name' column.
5. Use Fill Series to generate working dates for the month (weekdays only).



# MODULE 3: Formatting & Organizing Data

## 3.1 Font Formatting

- Bold: Ctrl+B | Italic: Ctrl+I | Underline: Ctrl+U
- Font Face — Use Calibri or Arial for professional reports. Avoid decorative fonts.
- Font Size — Standard: 11–12pt for body, 14–16pt for headers.
- Font Color — Use dark colors for readability. Avoid light grey text on white backgrounds.
- Strikethrough — Useful for marking completed tasks or deprecated items.

## 3.2 Number Formatting

Number formatting controls how numbers are displayed without changing the actual value stored in the cell. Access it via Ctrl+1 or the Number group in the Home tab.

| Format Type     | Display Example          | How to Apply                               |
|-----------------|--------------------------|--------------------------------------------|
| General         | 1234.5                   | Default — no specific format               |
| Number          | 1,234.50                 | Home → Number → ',' button or Ctrl+Shift+1 |
| Currency        | RM 1,234.50              | Home → Number dropdown → Currency          |
| Percentage      | 85.50%                   | Ctrl+Shift+5 or Home → %                   |
| Date (Short)    | 01/01/2024               | Ctrl+Shift+3 or Custom: DD/MM/YYYY         |
| Date (Long)     | January 1, 2024          | Custom format: MMMM D, YYYY                |
| Time            | 14:30:00                 | Custom format: HH:MM:SS                    |
| Custom — Colour | [Blue]#,###0;[Red]-#,##0 | Ctrl+1 → Custom → type format code         |

 **KEY INSIGHT:** The actual value stored in a cell NEVER changes when you apply number formatting. RM 1,234.50 and 1234.5 are the same underlying number — only the display changes.

## 3.3 Alignment & Merge

- Horizontal Alignment — Left (text), Center (headers), Right (numbers). Use Home → Alignment group.
- Vertical Alignment — Top, Middle, Bottom. Access via Home → Alignment or Ctrl+1.
- Wrap Text (Alt+H, W) — Forces text to wrap within the cell so tall rows show full content.
- Merge & Center — Combines multiple cells into one. Use for report titles ONLY — never in data tables.
- Indent — Use the Indent buttons to create visual hierarchy within a cell (great for subtotals).

 **NOTE:** Avoid using Merge & Center inside data tables. It breaks sorting, filtering, and VLOOKUP. Instead, use 'Center Across Selection' (Ctrl+1 → Alignment → Horizontal → Center Across Selection).

### 3.4 Borders & Cell Styles

- Apply borders via Home → Borders dropdown. Use 'Thick Box Border' to outline tables, 'Bottom Border' for totals rows.
- Cell Styles (Home → Cell Styles) — Pre-built styles for headings, totals, input cells, and output cells. Great for quick professional formatting.

 [Screenshot: Cell Styles gallery — Heading 1, Total, Input, Output styles]

### 3.5 Conditional Formatting

Conditional Formatting automatically applies formatting (colours, icons, data bars) to cells based on their values. Access it via Home → Conditional Formatting.

#### Common Rules:

- Highlight Cell Rules — Greater Than, Less Than, Between, Equal To, Duplicate Values
- Top/Bottom Rules — Top 10, Bottom 10, Above Average, Below Average
- Data Bars — Horizontal bars within cells proportional to value — great for quick comparison
- Color Scales — Gradient color (e.g., red → yellow → green) across a range
- Icon Sets — Traffic lights, arrows, checkmarks based on value thresholds

#### How to Apply a Highlight Rule:

6. Select the range (e.g., C2:C50 for Sales column)
7. Home → Conditional Formatting → Highlight Cells Rules
8. Select your condition (e.g., Greater Than)
9. Type the threshold value (e.g., 20000)
10. Choose a format (e.g., Green Fill with Dark Green Text)
11. Click OK

 [Screenshot: Conditional Formatting — Color Scale applied to a sales column]

### EXERCISE: Module 3 — Format a Professional Sales Report

1. Open the sample 'SalesReport.xlsx' file provided.
2. Apply currency format (RM) to all sales columns with 2 decimal places.
3. Format the header row: Dark Blue fill (1B3A6B), White Bold text, 12pt Arial.
4. Apply alternating row shading: Light Blue (D6E4F7) for even rows, White for odd rows.
5. Add Conditional Formatting to the 'Total Sales' column: Red for sales below RM5,000, Green for sales above RM20,000.
6. Apply a Bottom Border to the totals row and a Thick Box Border around the entire table.

## MODULE 4: Essential Formulas & Functions

### 4.1 Understanding Formulas

In Excel, every formula starts with an equals sign (=). Without the =, Excel treats your entry as plain text and will NOT calculate it.

| Term     | Explanation                      | Example                                 |
|----------|----------------------------------|-----------------------------------------|
| Formula  | Any calculation starting with =  | =100+50                                 |
| Function | A named formula built into Excel | =SUM(A1:A10)                            |
| Argument | The input(s) a function needs    | In =SUM(A1:A10), A1:A10 is the argument |
| Range    | A group of cells                 | A1:A10 = cells A1 through A10           |
| Operator | Mathematical symbols             | + – * / ^ (power)                       |

### 4.2 Cell References

Understanding cell references is the single most important concept in Excel. It determines what happens when you copy a formula to other cells.

#### Relative Reference (e.g., A1)

The reference changes relative to where you paste the formula. This is the default behavior and what you want most of the time.

`=B2+C2` → When copied to the next row, becomes `=B3+C3`

#### Absolute Reference (e.g., \$A\$1)

The reference stays fixed regardless of where you copy the formula. Use this for fixed values like tax rates, targets, or constants.

`=B2*$F$1` → `$F$1` stays fixed; B2 changes as you copy down

#### Mixed Reference (e.g., \$A1 or A\$1)

Either the row OR the column is fixed, but not both. Useful in multiplication tables.

`=$A2*B$1` → Column A fixed, Row 1 fixed – use in a times table

 **SHORTCUT:** Press F4 while typing or editing a cell reference to cycle through: A1 → \$A\$1 → A\$1 → \$A1. This is one of the most important Excel shortcuts to learn.

### 4.3 Core Functions Reference

| Function | Syntax      | Example & Result                   |
|----------|-------------|------------------------------------|
| =SUM ( ) | =SUM(range) | =SUM(B2:B10) → Total of all values |

| Function        | Syntax                  | Example & Result                            |
|-----------------|-------------------------|---------------------------------------------|
| =AVERAGE ( )    | =AVERAGE(range)         | =AVERAGE(C2:C10) → Mean value               |
| =MAX ( )        | =MAX(range)             | =MAX(D2:D50) → Highest value                |
| =MIN ( )        | =MIN(range)             | =MIN(D2:D50) → Lowest value                 |
| =COUNT ( )      | =COUNT(range)           | =COUNT(A2:A100) → Count of numeric cells    |
| =COUNTA ( )     | =COUNTA(range)          | =COUNTA(A2:A100) → Count of non-empty cells |
| =COUNTBLANK ( ) | =COUNTBLANK(range)      | =COUNTBLANK(A2:A100) → Count of empty cells |
| =TODAY ( )      | =TODAY()                | Returns today's date (updates daily)        |
| =NOW ( )        | =NOW()                  | Returns current date AND time               |
| =ROUND ( )      | =ROUND(number, digits)  | =ROUND(3.567, 2) → 3.57                     |
| =INT ( )        | =INT(number)            | =INT(3.9) → 3 (removes decimal)             |
| =ABS ( )        | =ABS(number)            | =ABS(-500) → 500 (always positive)          |
| =MOD ( )        | =MOD(dividend, divisor) | =MOD(10,3) → 1 (remainder)                  |
| =POWER ( )      | =POWER(number, power)   | =POWER(2,8) → 256                           |

## 4.4 Working with Dates

Excel stores dates as serial numbers (e.g., January 1, 1900 = 1). This means you can do arithmetic with dates.

- =TODAY()-B2 → Calculate number of days since a past date (e.g., age of an invoice, years of service)
- =DATEDIF(start\_date, end\_date, "Y") → Years between two dates
- =YEAR(date) → Extract year from a date | =MONTH(date) → Month | =DAY(date) → Day
- =EDATE(date, months) → Add months to a date (e.g., calculate contract end date)
- =NETWORKDAYS(start, end) → Count working days between two dates (excludes weekends)

### EXERCISE: Module 4 — Employee Salary Calculation

1. Open 'SalaryCalc.xlsx'. In column E, calculate Total Salary = Basic + Allowance + Bonus using a formula.
2. In column F, calculate Years of Service: =INT((TODAY()-D2)/365) where D2 is Date Joined.
3. At the bottom, use =SUM() to calculate Department Total Salary.
4. Use =AVERAGE() to find Average Salary across all employees.
5. Use =MAX() and =MIN() to find the highest and lowest salaries.
6. Use =COUNTA() to count the total number of employees in the list.

7. In a separate cell, set up a Tax Rate (e.g., 10%). Use Absolute Reference (\$G\$1) so all salary tax calculations refer to that one fixed cell.

## MODULE 5: Productivity Features

### 5.1 Freeze Panes

Freeze Panes locks rows or columns so they remain visible when you scroll through a large dataset.

| Option              | What It Does                                       | How to Apply                                                         |
|---------------------|----------------------------------------------------|----------------------------------------------------------------------|
| Freeze Top Row      | Locks Row 1 (header row) while scrolling down      | View → Freeze Panes → Freeze Top Row                                 |
| Freeze First Column | Locks Column A while scrolling right               | View → Freeze Panes → Freeze First Column                            |
| Freeze Custom Panes | Locks rows above and columns left of selected cell | Click cell B2 (for example), then View → Freeze Panes → Freeze Panes |
| Unfreeze            | Removes all frozen panes                           | View → Freeze Panes → Unfreeze Panes                                 |

 **BEST PRACTICE:** To freeze both the header row and the first column, click cell B2, then apply Freeze Panes. This freezes Row 1 and Column A simultaneously.

### 5.2 Split Window

Split Window divides the spreadsheet into two or four independent panes, each scrollable separately. This lets you compare different parts of the same worksheet.

- Click the cell where you want the split to occur
- Go to View → Split
- You'll see two or four scrollable panels of the same sheet
- To remove: View → Split again (toggle off)

### 5.3 Find & Replace

#### Find (Ctrl + F):

- Search for a value, text, or formula across the worksheet or entire workbook
- Use 'Match entire cell contents' for exact matches
- Use 'Look in: Formulas' to search within formula text

#### Find & Replace (Ctrl + H):

- Replace specific values with new ones across the worksheet or workbook
- Example: Replace all instances of 'KL' with 'Kuala Lumpur'
- Use 'Replace All' to change everything at once — review first!

 **NOTE:** Before using 'Replace All', always use 'Find All' first to preview what will be changed. Replacing incorrectly can corrupt your data.

### 5.4 Sorting Data

Sorting reorganizes your data alphabetically, numerically, or by date. Always convert your data to a Table (Ctrl+T) before sorting to protect data integrity.

### Simple Sort:

12. Click any cell within your data
13. Go to Data → Sort A to Z (or Z to A)
14. Or use the sort buttons in the Home → Editing group

### Multi-Level Sort:

15. Data → Sort (opens the full dialog)
16. Set Sort By (e.g., Department), then Add Level
17. Set Then By (e.g., Salary — Largest to Smallest)
18. Click OK

 **SORT SAFELY:** Always click inside your data range (not the header) before sorting. If Excel asks 'Expand the selection?' — always choose Expand. Sorting without expansion can mix up rows and misalign your data.

## 5.5 Filtering Data

AutoFilter lets you show only the rows that meet specific criteria, hiding everything else temporarily. Your data is not deleted — just hidden.

### Turn AutoFilter On/Off:

- Shortcut: Ctrl + Shift + L
- Or: Home → Sort & Filter → Filter

### Using Filters:

- Click the dropdown arrow in a column header
- Text filters: Contains, Begins With, Ends With
- Number filters: Greater Than, Less Than, Between, Top 10
- Date filters: This Month, Last Month, This Quarter, Custom Range
- Filter by Color — if you have applied conditional formatting or cell colours

### Clear Filters:

- Data → Clear (removes filter from all columns)
- Or click the filter dropdown in a specific column and choose 'Clear Filter From...'



### EXERCISE: Module 5 — Employee Database Navigation

1. Open 'EmployeeDB.xlsx'. Freeze Row 1 (the header) and Column A (Employee ID).
2. Apply AutoFilter (Ctrl+Shift+L). Filter the Department column to show only 'Sales' department employees.
3. Clear the filter and apply a new filter: Salary > RM 5,000.
4. Sort the data by Department (A to Z), then within each department by Salary (Largest to Smallest).
5. Use Find & Replace (Ctrl+H) to replace 'KL' with 'Kuala Lumpur' throughout the sheet.

## DAY 2 — MODULE 6: Intermediate Functions

### 6.1 IF Function

The IF function tests a condition and returns one value if TRUE and another if FALSE. It is the most widely used logical function in Excel.

```
=IF(logical_test, value_if_true, value_if_false)
```

| Example                  | Formula                               | Result                              |
|--------------------------|---------------------------------------|-------------------------------------|
| Pass/Fail based on score | =IF(B2>=50, "PASS", "FAIL")           | 'PASS' if score ≥ 50, else 'FAIL'   |
| Bonus calculation        | =IF(C2>5000, C2*0.05, 0)              | 5% bonus if salary > 5000, else 0   |
| Overdue check            | =IF(D2<TODAY(), "Overdue", "On Time") | 'Overdue' if date has passed        |
| Blank check              | =IF(A2="", "No Data", A2)             | Shows 'No Data' if cell A2 is empty |

### 6.2 Nested IF

Nested IF places one IF function inside another to handle multiple conditions (more than two outcomes).

```
=IF(D2>=90, "A", IF(D2>=80, "B", IF(D2>=70, "C", IF(D2>=60, "D", "F"))))
```

**💡 EXCEL 365 ALTERNATIVE:** Use the IFS() function instead of nested IFs — it is cleaner and easier to read: =IFS(D2>=90,"A", D2>=80,"B", D2>=70,"C", D2>=60,"D", TRUE,"F")

### 6.3 COUNTIF — Count with One Condition

```
=COUNTIF(range, criteria)
```

- =COUNTIF(B2:B50, "Sales") → Count rows where department is 'Sales'
- =COUNTIF(C2:C50, ">5000") → Count values greater than 5000
- =COUNTIF(D2:D50, "KL") → Count rows where city is 'KL'
- =COUNTIF(A2:A100, "\*Ahmad\*") → Count cells containing 'Ahmad' (use \* as wildcard)

### 6.4 SUMIF — Sum with One Condition

```
=SUMIF(range, criteria, sum_range)
```

- =SUMIF(B:B, "Sales", C:C) → Sum Column C only where Column B = 'Sales'
- =SUMIF(D:D, "KL", E:E) → Sum Column E only where Column D = 'KL'
- =SUMIF(C:C, ">10000", F:F) → Sum Column F only where Column C > 10000

### 6.5 AVERAGEIF — Average with One Condition

```
=AVERAGEIF(range, criteria, average_range)
```

- =AVERAGEIF(B:B, "HR", C:C) → Average of Column C only for HR department
- =AVERAGEIF(D:D, ">=80", E:E) → Average of Column E only where Column D ≥ 80



**NOTE:** COUNTIFS, SUMIFS, and AVERAGEIFS (with an 'S') support MULTIPLE conditions. Example: =SUMIFS(E:E, B:B, "Sales", D:D, "KL") — sums Column E where Department is 'Sales' AND City is 'KL'.

## 6.6 Text Functions

| Function       | Syntax                              | Example → Result                         |
|----------------|-------------------------------------|------------------------------------------|
| =LEFT ()       | =LEFT(text, num_chars)              | =LEFT("Ahmad Faizal",5) → "Ahmad"        |
| =RIGHT ()      | =RIGHT(text, num_chars)             | =RIGHT("EMP-001",3) → "001"              |
| =MID ()        | =MID(text, start, num_chars)        | =MID("880101-14-1234",7,2) → "14"        |
| =LEN ()        | =LEN(text)                          | =LEN("Hello") → 5                        |
| =UPPER ()      | =UPPER(text)                        | =UPPER("kuala lumpur") → "KUALA LUMPUR"  |
| =LOWER ()      | =LOWER(text)                        | =LOWER("KUALA LUMPUR") → "kuala lumpur"  |
| =PROPER ()     | =PROPER(text)                       | =PROPER("ahmad faizal") → "Ahmad Faizal" |
| =TRIM ()       | =TRIM(text)                         | =TRIM(" Hello ") → "Hello"               |
| =CONCAT ()     | =CONCAT(text1, text2, ...)          | =CONCAT(A2," ",B2) → "Ahmad Faizal"      |
| =TEXTJOIN ()   | =TEXTJOIN(delim,ignore_empty,range) | =TEXTJOIN(" ",TRUE,A2:A5)                |
| =TEXT ()       | =TEXT(value, format_text)           | =TEXT(TODAY(),"DD/MM/YYYY")              |
| =SUBSTITUTE () | =SUBSTITUTE(text,old,new)           | =SUBSTITUTE(A2,"-","") — remove hyphens  |

## 6.7 IFERROR — Handle Formula Errors Gracefully

=IFERROR(value, value\_if\_error)

- =IFERROR(A2/B2, 0) → Returns 0 instead of #DIV/0! when dividing by zero
- =IFERROR(VLOOKUP(A2,Table,2,FALSE), "Not Found") → Shows 'Not Found' instead of #N/A

| Excel Error | Meaning          | Common Cause                                      |
|-------------|------------------|---------------------------------------------------|
| #DIV/0!     | Division by zero | Dividing by a cell that is blank or contains zero |
| #N/A        | Value not found  | VLOOKUP lookup value does not exist in the table  |

| Excel Error | Meaning           | Common Cause                                       |
|-------------|-------------------|----------------------------------------------------|
| #VALUE!     | Wrong data type   | Math operation on a text cell                      |
| #REF!       | Invalid reference | Formula pointing to deleted or moved cells         |
| #NAME?      | Unrecognised name | Misspelled function name or undefined range name   |
| #NUM!       | Invalid number    | SQRT of a negative number, or number too large     |
| ####        | Column too narrow | Widen the column — data is there, just not visible |

**EXERCISE: Module 6 — Employee KPI Dashboard**

1. In column E, use IF to flag employees as 'Achieved' (KPI score > 80) or 'Not Achieved'.
2. In column F, use IFS (or nested IF) to assign Grade: Excellent (>90), Good (75-90), Satisfactory (60-74), Needs Improvement (<60).
3. Count employees in each department using COUNTIF.
4. Calculate average KPI score by department using AVERAGEIF.
5. Use IFERROR around a VLOOKUP to return 'Employee Not Found' instead of #N/A.
6. Use LEFT, MID, and PROPER functions to extract and clean employee name data.

# MODULE 7: Lookup Functions

## 7.1 VLOOKUP — Vertical Lookup

VLOOKUP searches for a value in the first column of a table and returns a value from a specified column in the same row. It is one of the most commonly used functions in Excel.

```
=VLOOKUP(lookup_value, table_array, col_index_num, [range_lookup])
```

| Argument      | Meaning                                                       | Example                                             |
|---------------|---------------------------------------------------------------|-----------------------------------------------------|
| lookup_value  | The value you are searching for                               | A2 (e.g., Employee ID or Product Code)              |
| table_array   | The range/table to search in                                  | ProductList!A:D (always start from leftmost column) |
| col_index_num | Column number to return (1 = first column of range)           | 3 = return value from 3rd column of the table       |
| range_lookup  | FALSE = Exact Match (use 99% of the time), TRUE = Approximate | FALSE                                               |

### VLOOKUP Example:

```
=VLOOKUP(A2, ProductList!$A:$D, 3, FALSE)
```

This formula looks up the value in A2 from the first column of ProductList!A:D, and returns the value from column 3 (Price).

### VLOOKUP Limitations to Know:

- Can only search in the LEFTMOST column of the table array — cannot look to the left
- Column index number must be hardcoded — fragile if columns are inserted/deleted
- Returns #N/A if the lookup value is not found — wrap with IFERROR
- Not available for searching to the left of the lookup column

## 7.2 HLOOKUP — Horizontal Lookup

HLOOKUP searches for a value in the first ROW of a table and returns a value from a specified row. Use it when your data is laid out horizontally.

```
=HLOOKUP(lookup_value, table_array, row_index_num, [range_lookup])
```

Same logic as VLOOKUP but searches across rows instead of columns.

## 7.3 XLOOKUP — The Modern Upgrade (Excel 365)

XLOOKUP replaces both VLOOKUP and HLOOKUP with a single, more powerful function. It is available exclusively in Excel 365 and Excel 2021.

```
=XLOOKUP(lookup_value, lookup_array, return_array, [if_not_found], [match_mode], [search_mode])
```

| Argument     | Meaning                                                                         |
|--------------|---------------------------------------------------------------------------------|
| lookup_value | The value you are looking for                                                   |
| lookup_array | The column (or row) to search in — does NOT need to be the first column         |
| return_array | The column (or row) to return values from                                       |
| if_not_found | Optional: what to display if not found (e.g., "Not Found" instead of #N/A)      |
| match_mode   | Optional: 0 = Exact (default), -1 = Next Smaller, 1 = Next Larger, 2 = Wildcard |
| search_mode  | Optional: 1 = First-to-last (default), -1 = Last-to-first, 2 = Binary Search    |

### XLOOKUP Example:

```
=XLOOKUP(A2, B:B, D:D, "Not Found")
```

Searches column B for the value in A2, returns the matching value from column D. Returns 'Not Found' if not matched.

| Feature               | VLOOKUP                 | XLOOKUP                                      |
|-----------------------|-------------------------|----------------------------------------------|
| Search Direction      | Left-to-Right only      | Any direction — left, right, up, down        |
| Column Identification | Column number (fragile) | Direct array reference (flexible)            |
| If Not Found          | Returns #N/A error      | Custom message via if_not_found argument     |
| Multiple Results      | One column only         | Can return a full row/column of values       |
| Horizontal Search     | Need HLOOKUP            | XLOOKUP handles both vertical and horizontal |
| Availability          | All Excel versions      | Excel 365 and Excel 2021 only                |

### EXERCISE: Module 7 — Customer & Product Lookup System

1. Build a Customer Lookup form: when you type a Customer ID in B2, cells B3:B7 show Name, Department, Email, Phone, and Region using VLOOKUP.
2. Wrap each VLOOKUP with IFERROR to show 'Customer Not Found' if the ID does not exist.
3. Create the same lookup using XLOOKUP and compare the formula simplicity.
4. Practice searching for the LAST occurrence of a value using XLOOKUP with search\_mode -1.

# MODULE 8: Excel Tables

## 8.1 What is an Excel Table?

An Excel Table is a structured range with a specific design and built-in behaviors that make managing data significantly easier. When you convert a range to a Table, Excel applies formatting, adds filter dropdowns, and enables powerful features.

## 8.2 Why Use Tables?

| Feature               | Benefit                                                                                                                            |
|-----------------------|------------------------------------------------------------------------------------------------------------------------------------|
| Auto-Expand           | When you add a new row or column, it is automatically included in all formulas, charts, and Pivot Tables referencing this Table.   |
| Structured References | Formulas use column names instead of cell addresses: =Table1[Sales] instead of =C2:C100. This is much easier to read and maintain. |
| Built-in AutoFilter   | Every column header gets a filter dropdown automatically.                                                                          |
| Total Row             | Click Table Design → Total Row to add a row with Sum, Average, Count, etc. for each column.                                        |
| Auto-Formatting       | Alternating row colors and header formatting are applied automatically.                                                            |
| Remove Duplicates     | Table Tools → Remove Duplicates — finds and removes duplicate rows instantly.                                                      |
| Slicer Support        | Tables can use Slicers for visual, interactive filtering — same as Pivot Tables.                                                   |

## 8.3 Creating a Table

- 19. Click any cell inside your data range
- 20. Press Ctrl + T (or go to Insert → Table)
- 21. Verify the range is correct in the dialog box
- 22. Make sure 'My table has headers' is checked if your data has a header row
- 23. Click OK
- 24. Optional: Go to Table Design tab → rename the table in the 'Table Name' box (e.g., 'SalesData')

 [Screenshot: Create Table dialog and the resulting formatted Table with filter dropdowns]

## 8.4 Structured References

Instead of writing =SUM(C2:C100), you can write =SUM(SalesData[Sales]) — which automatically adjusts when rows are added.

```
=SUM(SalesData[Total Sales])  
=AVERAGE(SalesData[KPI Score])  
=COUNTIF(SalesData[Department], "HR")
```

## 8.5 Remove Duplicates

25. Click inside the Table
26. Go to Table Design tab → Remove Duplicates
27. Select which columns to check for duplicates (usually check all)
28. Click OK — Excel will tell you how many duplicate rows were removed

 **IMPORTANT:** Remove Duplicates permanently deletes rows. Always create a backup copy of your worksheet before running this function.

### EXERCISE: Module 8 — Customer Database Cleaning

1. Open 'CustomerDB.xlsx'. Convert the data range to an Excel Table (Ctrl+T). Name it 'CustomerTable'.
2. Add a Total Row. Set the 'Revenue' column to show Sum, and the 'Region' column to show Count.
3. Use the filter dropdowns to show only customers from 'Kuala Lumpur'.
4. Run Remove Duplicates on the Customer ID column to find any duplicate entries.
5. Add a new column called 'Status' with the formula =IF(CustomerTable[Revenue]>10000, "Premium", "Standard").
6. Observe how the formula automatically fills down for all rows.

## MODULE 9: Pivot Tables & Pivot Charts

### 9.1 What is a Pivot Table?

A Pivot Table is a powerful tool that lets you summarize, analyze, and explore large datasets dynamically. Without writing a single formula, you can instantly calculate totals, averages, counts, and percentages — and then reorganize the view with drag and drop.

Think of it as an interactive summary table where you decide how to group, filter, and calculate your data — and Excel does all the work.

### 9.2 Creating Your First Pivot Table

29. Click any cell inside your data (convert to Table first — Ctrl+T — for best results)
30. Go to Insert → PivotTable
31. Choose 'New Worksheet' (recommended for clarity)
32. Click OK — the PivotTable Field List appears on the right
33. Drag fields from the Field List into the four areas: Rows, Columns, Values, and Filters

 [Screenshot: Pivot Table Field List — showing Rows, Columns, Values, Filters areas]

### 9.3 The Four Pivot Table Areas

| Area    | What Goes Here  | Effect                                     | Example                                |
|---------|-----------------|--------------------------------------------|----------------------------------------|
| ROWS    | Category fields | Each unique value becomes a row            | Department, Region, Product Name       |
| COLUMNS | Category fields | Each unique value becomes a column header  | Quarter, Month, Year                   |
| VALUES  | Numeric fields  | Calculates Sum, Count, Average, etc.       | Sales Amount, KPI Score, Units Sold    |
| FILTERS | Category fields | Adds a report filter above the Pivot Table | Year, Product Category, Employee Grade |

### 9.4 Changing the Value Calculation

34. Right-click any number inside the Values area of the Pivot Table
35. Select 'Value Field Settings'
36. Choose the calculation type: Sum, Count, Average, Max, Min, Product, Distinct Count
37. Click OK

 **SHOW AS %:** In Value Field Settings, click the 'Show Values As' tab. Choose '% of Grand Total' to show each value as a percentage of the total — great for market share analysis.

### 9.5 Filtering & Grouping

## Date Grouping:

- Right-click a date field in the Pivot Table → Group → select: Years, Quarters, Months
- This turns a field of individual dates into Year → Quarter → Month hierarchy

## Number Grouping:

- Right-click a number field → Group → set Starting at, Ending at, and By value
- Example: Group salary ranges into buckets: 0-3000, 3001-6000, 6001-10000+

## 9.6 Slicers

Slicers are visual filter buttons that sit on top of your Pivot Table, making filtering interactive and easy to use — especially in dashboard presentations.

38. Click inside the Pivot Table
39. Go to PivotTable Analyze → Insert Slicer
40. Tick the fields you want as Slicers (e.g., Region, Quarter, Category)
41. Click OK — Slicers appear on the sheet
42. Click any button on the Slicer to filter the Pivot Table
43. Hold Ctrl to select multiple values in a Slicer
44. Click the X button on the Slicer to clear the filter

 **CONNECT TO MULTIPLE PIVOT TABLES:** Right-click a Slicer → Report Connections. Link it to multiple Pivot Tables so one click filters everything simultaneously — perfect for dashboards.

## 9.7 Pivot Charts

A Pivot Chart is a chart linked to a Pivot Table. It updates automatically when the Pivot Table changes.

45. Click inside your Pivot Table
46. Go to PivotTable Analyze → PivotChart
47. Select your chart type (Bar, Column, Line, Pie, etc.)
48. Click OK
49. Format the chart using the Chart Design and Format tabs

## 9.8 Refreshing a Pivot Table

- When your source data changes, the Pivot Table does NOT update automatically
- Right-click inside the Pivot Table → Refresh
- Or: PivotTable Analyze → Refresh
- To auto-refresh when file opens: PivotTable Analyze → PivotTable Options → Data tab → 'Refresh data when opening the file'

## EXERCISE: Module 9 — Sales Analysis Dashboard

1. Open 'SalesData.xlsx'. Create a Pivot Table on a new sheet.



2. Set Rows = Region, Columns = Quarter (group dates by quarter), Values = Total Sales (Sum).
3. Add a second Values field: Total Sales, but set it to show as '% of Grand Total'.
4. Add a Slicer for Product Category. Test filtering by clicking different categories.
5. Create a Pivot Chart from the Pivot Table — choose a clustered Column chart.
6. Connect the Slicer to both the Pivot Table and the Pivot Chart.

# MODULE 10: Data Visualization & Charts

## 10.1 Choosing the Right Chart

| Chart Type        | Best Used When...                                      | Workplace Example           |
|-------------------|--------------------------------------------------------|-----------------------------|
| Column Chart      | Comparing values across categories side by side        | Monthly sales by branch     |
| Bar Chart         | Same as column but horizontal — better for long labels | Top 10 customers by revenue |
| Line Chart        | Showing trends over time                               | Revenue growth Jan–Dec      |
| Pie / Donut Chart | Showing parts of a whole (use 5–6 segments maximum)    | Market share by segment     |
| Area Chart        | Like a line chart but emphasizes cumulative volume     | Cumulative sales over time  |
| Scatter Chart     | Showing correlation between two numeric variables      | Sales vs. Marketing Spend   |
| Combo Chart       | Combining two different chart types on one axis        | Sales (Bar) + Target (Line) |

## 10.2 Creating a Chart

- 50. Select your data range (include headers)
- 51. Go to Insert → Charts group → select your chart type
- 52. Or press Alt + F1 to insert a default chart instantly
- 53. The chart appears on the same sheet — drag to reposition
- 54. Use the Chart Design and Format tabs to customize

 [Screenshot: Column Chart creation — select data, Insert → Chart, formatted result]

## 10.3 Chart Formatting Best Practices

### What to DO:

- Use a descriptive chart title — 'Monthly Sales Performance 2024' (not just 'Chart 1')
- Add data labels so readers can see exact values without a legend
- Remove gridlines (or use very light gridlines) for a clean, modern look
- Match your chart colors to your company brand or the theme of the training material
- Use a consistent color for the same data series across all charts in a report
- Add axis titles to explain what the axes represent

### What NOT to do:

- Do NOT use 3D charts — they distort data perception and look dated
- Do NOT use Pie charts with more than 6–7 slices — use a Bar chart instead
- Do NOT rely on color alone to distinguish data — use patterns or labels too

- Do NOT leave the default 'Chart Title' placeholder — always replace it

## 10.4 Chart Elements

| Element     | Where to Add/Edit                                                               |
|-------------|---------------------------------------------------------------------------------|
| Chart Title | Click '+ Chart Elements' button → Chart Title. Double-click to edit text.       |
| Axis Titles | Chart Elements → Axis Titles. Click each to edit.                               |
| Data Labels | Chart Elements → Data Labels. Right-click → Format Data Labels to position.     |
| Legend      | Chart Elements → Legend. Move it to bottom, right, or remove if unnecessary.    |
| Gridlines   | Chart Elements → Gridlines. Uncheck to remove, or format for subtle lines.      |
| Error Bars  | Chart Elements → Error Bars. Use to show margin of error in scientific data.    |
| Trendline   | Chart Elements → Trendline. Adds a Linear, Exponential, or Moving Average line. |

## 10.5 Combo Chart (Bar + Line)

Combo charts are particularly useful for showing actual values alongside a target, budget, or average line.

55. Select your data (include both series)
56. Insert → Insert Combo Chart → Custom Combination
57. Assign one series as 'Clustered Column' and the other as 'Line'
58. Optionally tick 'Secondary Axis' for the Line series if scales are very different
59. Click OK

### EXERCISE: Module 10 — Sales Dashboard Visualization

1. From the Sales dataset, create a Column Chart showing Monthly Total Sales.
2. Format: Remove gridlines, add data labels, change bar color to dark blue (1B3A6B), add a descriptive title.
3. Create a Pie Chart showing Sales by Region. Format: Add % labels, remove legend if using data labels.
4. Create a Combo Chart: Actual Sales (Column) + Target (Line). Add a secondary axis for the Target line.
5. Arrange all three charts on a new 'Dashboard' sheet with a title banner.

## MODULE 11: Print Settings & Report Preparation

### 11.1 Print Area

Setting a Print Area tells Excel exactly which part of the worksheet to print, ignoring everything else.

60. Select the cells you want to print
61. Go to Page Layout → Print Area → Set Print Area
62. A dashed border appears around the selected range
63. To clear: Page Layout → Print Area → Clear Print Area

### 11.2 Page Setup Options

| Setting         | Where to Find It                                | When to Use                                                                  |
|-----------------|-------------------------------------------------|------------------------------------------------------------------------------|
| Orientation     | Page Layout → Orientation                       | Landscape for wide tables, Portrait for lists                                |
| Paper Size      | Page Layout → Size                              | A4 (default in Malaysia), or Letter                                          |
| Margins         | Page Layout → Margins                           | Narrow margins for data-heavy reports                                        |
| Scale to Fit    | Page Layout → Scale to Fit                      | Set Width to '1 Page' to prevent the last column printing on a separate page |
| Print Gridlines | Page Layout → Sheet Options → Print (Gridlines) | Useful for data entry forms                                                  |
| Print Headings  | Page Layout → Sheet Options → Print (Headings)  | Shows A, B, C row/column labels in printout                                  |

### 11.3 Headers & Footers

64. Go to Insert → Header & Footer
65. Click in the Left, Center, or Right section of the header/footer
66. Type your text, or use the Header & Footer buttons to insert: Page Number, Total Pages, Date, Time, File Name, Sheet Name
67. Click outside the header area when done

#### Useful Footer Codes:

|                          |  |                  |  |                   |
|--------------------------|--|------------------|--|-------------------|
| &P = Current Page Number |  | &N = Total Pages |  | &D = Today's Date |
| &F = File Name           |  | &A = Sheet Name  |  | &T = Current Time |

 **PROFESSIONAL FOOTER:** Use: &"Arial,Bold"Confidential — &P of &N — Printed: &D  
This creates a footer like: 'Confidential — Page 1 of 5 — Printed: 25/05/2026'

### 11.4 Repeat Rows on Every Printed Page

For long tables, repeat the header row at the top of each printed page so readers always know what each column means.

68. Page Layout → Print Titles

69. In the 'Rows to repeat at top' box, click the row selector for Row 1 (shows as \$1:\$1)

70. Click OK

## 11.5 Page Breaks

- Insert Manual Page Break — Click on the row where you want the break, then Page Layout → Breaks → Insert Page Break
- View Page Breaks — View → Page Break Preview. Drag the blue lines to adjust where pages split.
- Remove All Page Breaks — Page Layout → Breaks → Reset All Page Breaks

## 11.6 Export to PDF

### Method 1 — File → Export:

71. File → Export → Create PDF/XPS

72. Choose location and file name

73. Click Publish

### Method 2 — File → Save As:

74. File → Save As → change file type to PDF

75. Click Options to set print area, specific pages, or document properties

 **PDF TIP:** Always preview your document before exporting to PDF. Use File → Print → Print Preview to see exactly how it will look. Adjust print area and scaling as needed before exporting.

# FINAL MINI PROJECT: Company Sales Performance Dashboard

## Scenario

You are a Data Analyst at Maju Berhad Sdn Bhd. The Sales Manager has given you a raw dataset of 500 sales transactions from 2023 and asked you to build a comprehensive Sales Performance Dashboard.

## Dataset Overview

The file 'SalesData\_Raw.xlsx' contains the following columns:

| Column         | Data Type | Description                                                         |
|----------------|-----------|---------------------------------------------------------------------|
| Transaction ID | Text      | Unique ID for each sale (e.g., TXN-001)                             |
| Date           | Date      | Date of transaction (may have inconsistent formats)                 |
| Salesperson    | Text      | Full name of the salesperson                                        |
| Region         | Text      | Region (KL, Selangor, Penang, Johor, Sabah)                         |
| Product Code   | Text      | Internal product code (look up Product Name from reference table)   |
| Category       | Text      | Product category (Electronics, Furniture, Stationery, IT)           |
| Quantity       | Number    | Units sold (some cells may be blank or contain errors)              |
| Unit Price     | Number    | Price per unit                                                      |
| Total Sales    | Blank     | To be calculated: Quantity × Unit Price                             |
| Commission     | Blank     | To be calculated using IF: 5% if Total Sales > 5000, else 3%        |
| Status         | Blank     | To be flagged: 'High Value' if Total Sales > 20000, else 'Standard' |

## Project Tasks

### TASK 1 — Clean the Dataset

- Use TRIM() to remove leading/trailing spaces from the Salesperson column
- Use PROPER() to fix inconsistent name casing
- Use SUBSTITUTE() to standardise Region names (e.g., replace 'K.L.' with 'KL')
- Check for and remove any completely blank rows using Find & Select → Go To Special → Blanks
- Ensure the Date column is formatted consistently as DD/MM/YYYY

### TASK 2 — Apply Formulas

- Calculate Total Sales in column I: =G2\*H2
- Calculate Commission using IF: =IF(I2>5000, I2\*0.05, I2\*0.03)
- Flag Status using IF: =IF(I2>20000, "High Value", "Standard")
- Use VLOOKUP or XLOOKUP to bring in Product Name from the reference sheet using Product Code
- Wrap all VLOOKUP/XLOOKUP formulas with IFERROR to handle missing product codes

### TASK 3 — Convert to Excel Table

- Select all data and press Ctrl+T to convert to a Table
- Name the table 'SalesData'
- Add a Total Row showing Sum for Total Sales and Commission
- Run Remove Duplicates on Transaction ID to check for any duplicate entries

### TASK 4 — Build a Pivot Table

- Create a Pivot Table on a new sheet called 'Analysis'
- Analysis 1 — Rows: Region | Columns: Quarter | Values: Total Sales (Sum)
- Analysis 2 — Rows: Salesperson | Values: Total Sales (Sum), Commission (Sum), Transaction Count
- Analysis 3 — Rows: Category | Values: Total Sales (Sum), as % of Grand Total
- Add Slicers for Region and Category

### TASK 5 — Create Dashboard Charts

- Monthly Sales Trend — Line chart showing Total Sales by month
- Regional Sales — Bar chart showing Total Sales by Region
- Category Share — Pie or Donut chart showing % of sales by Category
- Salesperson Performance — Horizontal Bar chart, top 5 by Total Sales

### TASK 6 — Design the Dashboard Page

- Create a new sheet called 'Dashboard'
- Add a title banner: 'Maju Berhad — Sales Performance Dashboard 2023'
- Place all four charts on this sheet, organized neatly
- Link the Slicers to all charts and Pivot Tables
- Format the dashboard with your chosen color scheme (dark blue + gold recommended)

### TASK 7 — Print & Export

- Set the Dashboard sheet as the Print Area
- Scale to fit on 1 page
- Add footer: Prepared by: [Your Name] | Page &P of &N | &D
- Export the entire workbook as PDF

## Assessment Rubric

| Task                         | Skills Assessed                                     | Marks |
|------------------------------|-----------------------------------------------------|-------|
| Data Cleaning                | TRIM, PROPER, SUBSTITUTE, Remove Blanks             | 15    |
| Formulas — Total, Commission | SUM, IF, Absolute Reference                         | 15    |
| VLOOKUP / XLOOKUP            | Correct syntax, col index, IFERROR                  | 15    |
| Excel Table Setup            | Structured reference, Total Row, Remove Duplicates  | 10    |
| Pivot Table Analysis         | 3 correct Pivot Tables, correct field placement     | 20    |
| Slicers & Interactivity      | Slicers connected to all Pivot Tables and charts    | 10    |
| Dashboard Charts             | 4 correct chart types, proper formatting and titles | 10    |
| Print & PDF Export           | Print area, footer, correct scaling                 | 5     |
| Total                        |                                                     | 100   |



QUICK REFERENCE CARD — Formulas & Shortcuts

Essential Formulas

| Function       | Syntax                         | Key Use                          |
|----------------|--------------------------------|----------------------------------|
| =SUM ( )       | =SUM(range)                    | Add all values                   |
| =AVERAGE ( )   | =AVERAGE(range)                | Mean value                       |
| =COUNT ( )     | =COUNT(range)                  | Count numbers                    |
| =COUNTA ( )    | =COUNTA(range)                 | Count non-empty                  |
| =MAX ( )       | =MAX(range)                    | Highest value                    |
| =MIN ( )       | =MIN(range)                    | Lowest value                     |
| =IF ( )        | =IF(test,true,false)           | Conditional logic                |
| =IFS ( )       | =IFS(test1,val1,...)           | Multiple conditions              |
| =COUNTIF ( )   | =COUNTIF(range,criteria)       | Count with 1 condition           |
| =SUMIF ( )     | =SUMIF(range,crit,sum_rng)     | Sum with 1 condition             |
| =AVERAGEIF ( ) | =AVERAGEIF(range,crit,avg_rng) | Average with 1 condition         |
| =COUNTIFS ( )  | =COUNTIFS(r1,c1,r2,c2,...)     | Count with multiple conditions   |
| =SUMIFS ( )    | =SUMIFS(sum,r1,c1,r2,c2,...)   | Sum with multiple conditions     |
| =VLOOKUP ( )   | =VLOOKUP(val,tbl,col,FALSE)    | Vertical lookup (classic)        |
| =XLOOKUP ( )   | =XLOOKUP(val,arr,ret,"?")      | Vertical/Horizontal lookup (365) |
| =IFERROR ( )   | =IFERROR(formula,"msg")        | Handle errors gracefully         |
| =LEFT ( )      | =LEFT(text,n)                  | Extract from left                |
| =RIGHT ( )     | =RIGHT(text,n)                 | Extract from right               |
| =MID ( )       | =MID(text,start,n)             | Extract from middle              |
| =TRIM ( )      | =TRIM(text)                    | Remove extra spaces              |
| =PROPER ( )    | =PROPER(text)                  | Capitalise each word             |
| =CONCAT ( )    | =CONCAT(a," ",b)               | Join text                        |
| =TEXTJOIN ( )  | =TEXTJOIN(" ",TRUE,range)      | Join with delimiter              |
| =TODAY ( )     | =TODAY()                       | Today's date                     |
| =NOW ( )       | =NOW()                         | Current date & time              |
| =ROUND ( )     | =ROUND(num,digits)             | Round a number                   |
| =TEXT ( )      | =TEXT(value,"DD/MM/YYYY")      | Format number as text            |

## Essential Keyboard Shortcuts

| Shortcut         | Action                | Shortcut         | Action             |
|------------------|-----------------------|------------------|--------------------|
| Ctrl + N         | New Workbook          | Ctrl + Z / Y     | Undo / Redo        |
| Ctrl + S         | Save                  | Ctrl + C / V / X | Copy / Paste / Cut |
| F12              | Save As               | Ctrl + F         | Find               |
| Ctrl + P         | Print                 | Ctrl + H         | Find & Replace     |
| Ctrl + Home      | Go to A1              | Ctrl + End       | Last used cell     |
| Ctrl + Shift + L | Toggle Filter         | F2               | Edit cell          |
| Ctrl + T         | Create Table          | Alt + Enter      | New line in cell   |
| Ctrl + 1         | Format Cells          | Ctrl + Shift + ~ | Show formulas      |
| F4               | Toggle \$\$ reference | Ctrl + D         | Fill Down          |
| Ctrl + E         | Flash Fill            | Shift + F11      | New worksheet      |
| Alt + H + O + I  | AutoFit column width  | Ctrl + Shift + + | Insert row/column  |
| Ctrl + -         | Delete row/column     | Ctrl + Shift + 1 | Number format      |
| Ctrl + Shift + 5 | Percentage format     | Ctrl + Shift + 3 | Date format        |

## Common Excel Errors

| Error   | Meaning                         | Fix                                               |
|---------|---------------------------------|---------------------------------------------------|
| #DIV/0! | Dividing by zero or blank       | Wrap with IFERROR, or check if denominator is 0   |
| #N/A    | Value not found (VLOOKUP)       | Wrap with IFERROR, check spelling of lookup value |
| #VALUE! | Wrong data type in formula      | Check for text in cells used in math formulas     |
| #REF!   | Formula points to deleted cells | Re-link the formula or undo the deletion          |
| #NAME?  | Function name misspelled        | Check spelling — Excel functions have no spaces   |
| #NUM!   | Invalid numeric operation       | Check for SQRT of negative, or overflow numbers   |
| ####    | Column too narrow to show value | Widen the column by double-clicking column border |

## Formatting Quick Guide

| Format            | Shortcut                             | Example Display  |
|-------------------|--------------------------------------|------------------|
| Currency          | Ctrl+Shift+4 (\$) or apply custom RM | RM 1,234.50      |
| Percentage        | Ctrl+Shift+5                         | 85.50%           |
| Date              | Ctrl+Shift+3                         | 01/01/2024       |
| Number with comma | Ctrl+Shift+1                         | 1,234.00         |
| Remove formatting | Ctrl+Shift+~                         | 1234.5 (general) |
| Bold              | Ctrl+B                               | Text             |
| Italic            | Ctrl+I                               | Text             |
| Underline         | Ctrl+U                               | Text             |